

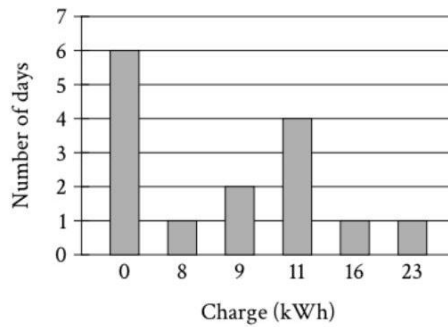
# SAT 7

## Math Module 3

Question # ID

SAT7 M 3.1 a07ma301

The bar graph summarizes the charge, in kilowatt-hours (kWh), a battery received each day for 15 days.



For how many of these 15 days did the battery receive a charge of 0 kWh?

(A) 0

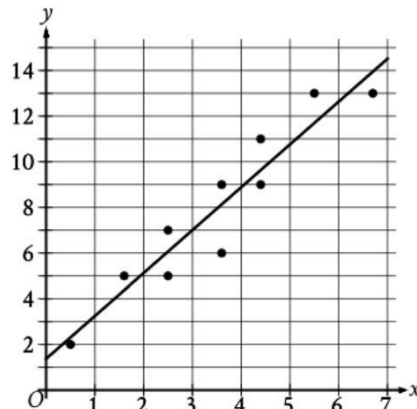
(B) 1

(C) 4

(D) 6

SAT7 M 3.2 a07ma302

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit shown?

(A) 0

(B)  $\frac{1}{2}$

(C) 1

(D) 2

SAT7 M 3.3 a07ma303

In a sample, 80% of the items are faulty. There are 88 faulty items in the sample. How many total items are in the sample?

## SAT 7

## Math Module 3

Question # ID

SAT7 M 3.4 a07ma304

For the exponential function  $g$ , the table shows four values of  $x$  and their corresponding values of  $g(x)$ . Which equation defines  $g$ ?

☐ (A)  $g(x) = -25^x$

☐ (B)  $g(x) = -\left(\frac{1}{25}\right)^x$

☐ (C)  $g(x) = 25^x$

☐ (D)  $g(x) = \left(\frac{1}{25}\right)^x$

| $x$ | $g(x)$          |
|-----|-----------------|
| -1  | 25              |
| 0   | 1               |
| 1   | $\frac{1}{25}$  |
| 2   | $\frac{1}{625}$ |

SAT7 M 3.5 a07ma305

If  $8x = 6$ , what is the value of  $72x$ ?

☐ (A) 3

☐ (B) 15

☐ (C) 54

☐ (D) 57

SAT7 M 3.6 a07ma306

Which expression is equivalent to  $\frac{h^{16}q^7}{h^8q^{21}}$ , where  $h > 0$  and  $q > 0$ ?

☐ (A)  $\frac{h^{10}}{q^{14}}$

☐ (B)  $\frac{h^3}{q^3}$

☐ (C)  $h^{10}q^{14}$

☐ (D)  $h^3q^3$

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Question # ID

SAT7 M 3.7 a07ma307

A company fills boxes with approximately 23 pounds of oranges. To test the accuracy of the filling process, 344 boxes of oranges were selected at random and weighed. Based on the sample, it is estimated that the average weight of all boxes of oranges filled by the company in an 8-hour period is 23.1 pounds, with an associated margin of error of 0.19 pounds. Which of the following is the best interpretation of this estimate?

- (A) Plausible values for the average weight of all boxes of oranges filled by the company are between 22.91 pounds and 23.29 pounds.
- (B) Plausible values for the average weight of all boxes of oranges filled by the company are less than 22.91 pounds or greater than 23.29 pounds.
- (C) The average weight of all boxes of oranges filled by the company is less than 23.01 pounds.
- (D) The average weight of all boxes of oranges filled by the company is greater than 23.01 pounds.

SAT7 M 3.8 a07ma308

$$\frac{1}{3}(x + 6) - \frac{1}{2}(x + 6) = -8$$

What value of  $x$  is the solution to the given equation?

## SAT 7

## Math Module 3

Question # ID  
SAT7 M 3.9 a07ma309

$$f(x) = x^2 + bx$$
$$g(x) = 9x^2 - 27x$$

Functions  $f$  and  $g$  are given, and in function  $f$ ,  $b$  is a constant. If  $f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$ , what is the value of  $b$ ?

(A)  $-26$

(B)  $-\frac{26}{9}$

(C)  $\frac{1}{9}$

(D)  $9$

SAT7 M 3.10 a07ma310

In the  $xy$ -plane, line  $s$  passes through the point  $(0, 0)$  and is parallel to the line represented by the equation  $y = 18x + 2$ . If line  $s$  also passes through the point  $(4, d)$ , what is the value of  $d$ ?

(A)  $2$

(B)  $18$

(C)  $72$

(D)  $74$

## SAT 7

## Math Module 3

Question #

ID

SAT7 M 3.11

a07ma311

$$f(x) = (x - 44)(x - 46)$$

The function  $f$  is defined by the given equation. For what value of  $x$  does  $f(x)$  reach its minimum?

☐ (A) 46☐ (B) 45☐ (C) 44☐ (D) -1

SAT7 M 3.12 a07ma312

In triangle  $XYZ$ , angle  $Y$  is a right angle, point  $P$  lies on  $\overline{XZ}$ , and point  $Q$  lies on  $\overline{YZ}$  such that  $\overline{PQ}$  is parallel to  $\overline{XY}$ . If the measure of angle  $XZY$  is  $63^\circ$ , what is the measure, in degrees, of angle  $XPQ$ ?

SAT7 M 3.13 a07ma313

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by  $x$ . The product of 12 and the fourth odd integer is at most 26 less than the sum of the first and third odd integers. Which inequality represents this situation?

☐ (A)  $12(x + 6) \leq x + (x + 4) - 26$ ☐ (B)  $12(x + 6) \geq 26 - (x + (x + 4))$ ☐ (C)  $12(x + 4) \leq x + (x + 3) - 26$ ☐ (D)  $12(x + 4) \geq 26 - (x + (x + 3))$

# SAT 7

## Math Module 3

Question # ID  
SAT7 M 3.14 a07ma314

$$\frac{2}{5}x + \frac{7}{5}y = \frac{2}{7}$$

$$gx + ky = \frac{5}{2}$$

In the given system of equations,  $g$  and  $k$  are constants. The system has infinitely many solutions. What is the value of  $\frac{g}{k}$ ?

SAT7 M 3.15 a07ma315

The table shows three values of  $x$  and their corresponding values of  $y$ , where  $s$  is a constant. There is a linear relationship between  $x$  and  $y$ . Which of the following equations represents this relationship?

(A)  $sx + 3y = 18s$

(B)  $3x + sy = 18s$

(C)  $3x + sy = 18$

(D)  $sx + 3y = 18$

| $x$   | $y$ |
|-------|-----|
| $-2s$ | 24  |
| $-s$  | 21  |
| $s$   | 15  |

SAT7 M 3.16 a07ma316

$$y + k = x + 26$$

$$y - k = x^2 - 5x$$

In the given system of equations,  $k$  is a constant. The system has exactly one distinct real solution. What is the value of  $k$ ?

# SAT 7

## Math Module 3

Question # ID

SAT7 M 3.17 a07ma317

The function  $f$  is defined by  $f(x) = \frac{1}{2}(x + 6)$ . What is the value of  $f(4)$ ?

(A) 20

(B) 12

(C) 10

(D) 5

SAT7 M 3.18 a07ma318

The area of a rectangular banner is **2,661** square inches. The banner's length  $x$ , in inches, is 24 inches longer than its width, in inches. Which equation represents this situation?

(A)  $0 = x^2 - 24x - 2,661$

(B)  $0 = x^2 - 24x + 2,661$

(C)  $0 = x^2 + 24x - 2,661$

(D)  $0 = x^2 + 24x + 2,661$

SAT7 M 3.19 a07ma319

A right circular cone has a volume of **71,148** $\pi$  cubic centimeters and the area of its base is **5,929** $\pi$  square centimeters. What is the slant height, in centimeters, of this cone?

(A) 12

(B) 36

(C) 77

(D) 85

## SAT 7

## Math Module 3

Question # ID  
SAT7 M 3.20 a07ma320

$$7x + 6y = 5$$

$$28x + 24y = 20$$

For each real number  $r$ , which of the following points lies on the graph of each equation in the  $xy$ -plane for the given system?

(A)  $\left(r, -\frac{6r}{7} + \frac{5}{7}\right)$

(B)  $\left(r, \frac{7r}{6} + \frac{5}{6}\right)$

(C)  $\left(\frac{r}{4} + 5, -\frac{r}{4} + 20\right)$

(D)  $\left(-\frac{6r}{7} + \frac{5}{7}, r\right)$

SAT7 M 3.21 a07ma321

The cost of renting a carpet cleaner is \$52 for the first day and \$26 for each additional day. Which of the following functions gives the cost  $C(d)$ , in dollars, of renting the carpet cleaner for  $d$  days, where  $d$  is a positive integer?

(A)  $C(d) = 26d + 26$

(B)  $C(d) = 26d + 52$

(C)  $C(d) = 52d - 26$

(D)  $C(d) = 52d + 78$



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## Math Module 3

Question # ID

SAT7 M 3.22 a07ma322

The speed of a vehicle is increasing at a rate of **7.3** meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth?  
(Use 1 mile = 1,609 meters.)

☐ (A) 0.3☐ (B) 16.3☐ (C) 195.8☐ (D) 220.4